

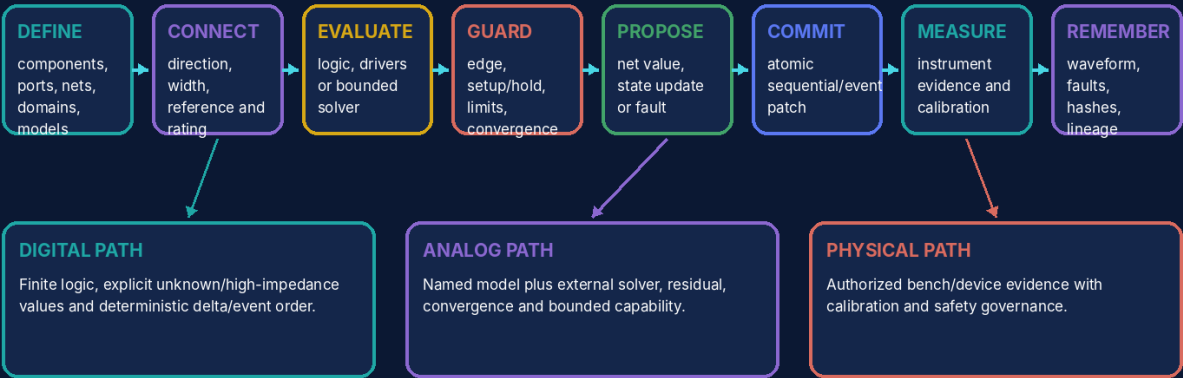
UNIVERSAL GEOMETRIC TOPOLOGICAL
OPEN MODULAR SUBSTRATE

UGTOMS VOLT 1.0

Typed Circuit, Logic, Timing, Model and Evidence Cartridge

UGTOMS VOLT 1.0 architecture

Ports, nets, values, clocks and faults become typed event-authoritative records.



A valid netlist or simulation result is not automatically a safe physical circuit.

Pure domain evaluation may propose evidence; only the retained authority chain may commit state.

ugtoms.domain.volt@1.0.0

Component	VOLT
Canonical identity	ugtoms.domain.volt@1.0.0
Component type	Domain cartridge
Version / date	1.0.0 / 30 August 2026
Document type	Open modular cartridge specification and implementation-ready profile
Status	Formal domain cartridge; specification self-checked; electrical design, safety and hardware validation not claimed

ATTRIBUTION AND EVIDENCE BOUNDARY

Prepared for Tom Klootwijk. Requester attribution is recorded as supplied and is not independently verified. This is a technical design artifact, not legal proof, scientific validation, regulatory approval or production safety certification.

Contents and Reading Rule

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2	Source basis and evidence discipline
3	Open modular cartridge ABI
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7	Clock domains and sequential event order
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READING RULE	
The UGTOMS source line supplies the reusable authority, typing, order, evidence and packing discipline. The specialist domain model in this cartridge is an engineering-derived extension. Where the source corpus is silent, this report says so instead of presenting general domain knowledge as source-derived.	

Executive Decision and Release Scope

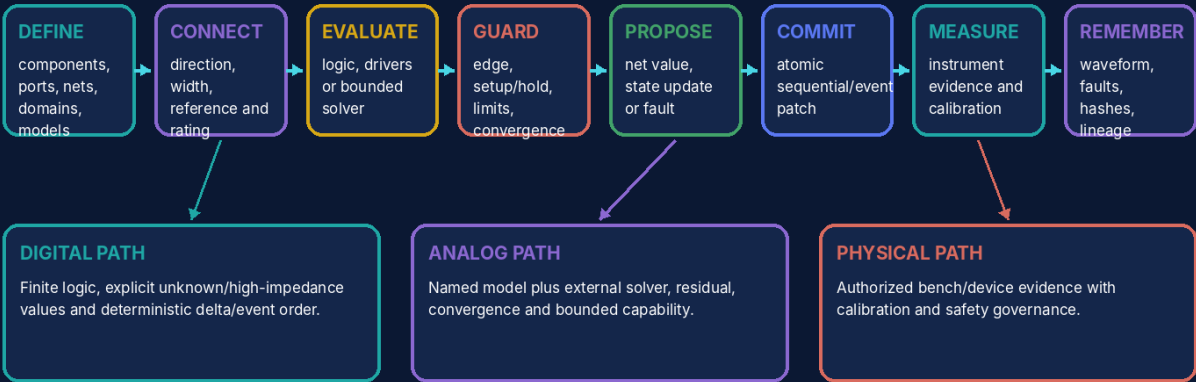
FORMAL DECISION
GO for VOLT as the circuit and logic cartridge. Connectivity, signal domains, finite logic, timing windows, sequential state, model residuals, faults, measurements and lineage become one typed substrate while mature HDL, SPICE, formal, PCB and laboratory tools remain external adapters.

VOLT maps especially cleanly onto the UGTOMS six-question chain: where a net or domain is relevant; whether ports and ratings match; whether an edge, threshold or convergence condition is ready; whether the proposal passed; which state changes; and which waveform/evidence must be remembered.

- Digital logic may be exact under a finite declared logic profile.
- Timing and analog results are bounded/model-scoped certificates rather than universal physical truth.
- A physical device or bench event commits only through authorization, calibration, evidence and safety gates.

UGTOMS VOLT 1.0 architecture

Ports, nets, values, clocks and faults become typed event-authoritative records.



A valid netlist or simulation result is not automatically a safe physical circuit.

Pure domain evaluation may propose evidence; only the retained authority chain may commit state.

Figure 1. VOLT keeps symbolic, model and physical authority paths distinct.

Source Basis and Evidence Discipline

The cartridge inherits the following UGTOMS commitments:

- Typed definitions and operators live inside the substrate, with stable IDs, dependencies and content hashes [S1-S3].
- Pure evaluation is upstream of support, compatibility, guard, verified proposal, deterministic commit and lineage [S1, S2, S4].
- Coordinates, pictures, sounds, meshes and packed codes are not identity or authority by themselves [S4-S8].
- Uncertainty, exactness, capability and error margins are explicit and may only be strengthened by evidence [S1, S2, S8].
- Component-scoped identities prevent a specialist domain pack from masquerading as a universal replacement of every other release [S10].

ENGINEERING-DERIVED DOMAIN CONTENT

The circuit type system, digital logic values, timing profiles, analog model adapters, fault records and examples are engineering-derived. The UGTOMS sources supply the reusable typed-operator, event, order, evidence and packing discipline; they do not supply a pre-existing electrical standard or component library.

ID	Registered source	Use in this cartridge	Pages
S1	UGTS__FOUNDATION__UNICODE_v5.pdf	Foundation 5.0 OperatorCells, literal/glyph/semantic separation, converse relations, exactness/capability and packed execution boundaries.	pp. 1-21
S2	UGTS_KC_4_2_General_Operator_Order_Addendum(1).pdf	Typed OperatorSpec, dependency and phase order, deterministic reduction, certificates, trace queries and open extension contracts.	pp. 2-32
S3	UGTS_KC_3_6_Tom_Klootwijk.pdf	Definitions as first-class content-addressed records, resolvable references, explicit operation order and deterministic trace.	pp. 1-15
S4	Unified_Geometric_Topological_Substrate_GPU_Native_Addendum.pdf	Canonical support → compatibility → guard → event → transition → lineage authority and narrow packed-state semantics.	pp. 2-15
S5	UGTS_KC_2_0_Tom_Klootwijk_Expanded_Substrate.pdf	Typed state, patterns, topology, bounded dynamics, hybrid events and numerical/error contracts.	pp. 3-15
S6	UGTS_KC_Two_Hands_3_0_Interactive_Graphics_Runtime.pdf	Stable identity, proposal verification, deterministic commit, checkpoints, rollback and replay.	pp. 3-16
S7	UGTS_KC_3_9_KC_Elizabeth_Vector_Game_Runtime.pdf	Versioned game projects, fixed-step runtime, snapshots/hashes, CLI and self-contained HTML delivery.	pp. 2-16
S8	UGTS_KC_4_0_Spatial_Evidence_Ledger(1).pdf	Observation versus authority, uncertainty, provenance, atomic patches, checkpoints and evidence-bearing lineage.	pp. 3-14
S9	UGTS_KC_3_6_2_SCLP_Tom_Klootwijk.pdf	Finite packing, interval guards, winding, bounded grammars and explicit topology/rotation distinctions.	pp. 1-15
S10	UGTS_Versioning_Charter_Phase_1_Foundation_Chronology(1).pdf	Component-scoped identities and explicit release graphs instead of one misleading global version ladder.	pp. 1-6
S11	UGTS_KC_Definition_Breaker_DB_0_1_Concept(1).pdf	Rules as typed physical objects, candidate preview, deterministic verification, commit, replay and reason-coded rejection.	pp. 2-24
S12	UGTS_KC_CHESS_1_0_Proof_Carrying_Solver(1).pdf	Exact rule authority, bounded proof certificates, heuristic separation and independently verifiable replay.	pp. 2-20
S13	UGTS_KC_4_2_GO_Solver_Report(1).pdf	History-correct exact state, finite minimax certificates, search/non-proof separation and self-contained delivery.	pp. 1-15

Open Modular Cartridge ABI

The cartridge is a concrete instance of the shared domain object D_VOLT:

D = (T, O, R, U, I, Q, E, G, Delta, L, Cert, K, Pi)	
Field	Contract
T	Entity, value and state types
O	Typed operators and transformations
R	Relations, graphs, topology and converse pairs
U	Units, dimensions, coordinate and profile conventions
I	Invariants, conservation laws and constraints
Q	Dependency, precedence, phase and reduction order
E	Observations, evidence, uncertainty and provenance
G	Guards, thresholds and admission predicates
Delta	Proposed and committed state patches
L	Lineage, replay, history and migration
Cert	Exactness, residual, interval, scope and status certificates
K	Cold atlas, hot codebook and packed representation
Pi	Optional visual, audio, physical or device projection

A conforming cartridge contains a manifest, type registry, operator registry, relation atlas, unit/profile system, dependency DAG, guard library, transition definitions, claims ledger, examples and validation fixtures. External mature libraries may sit behind adapters when they are safer or more accurate than a fresh reimplementation.

General Operator Record and Order

OperatorSpec = (id, version, syntax, arity, domains, codomain, units, shape, laws, exactness, effects, capabilities, dependencies, order_contract, invariants, provenance, content_hash)
--

The record follows the General Operator and Order layer [S2]. A symbol or glyph is not allowed to smuggle in undeclared type, unit, order, inverse or authority semantics.

EXAMPLE RECORD
volt.sequential.capture consumes a ClockEdgeProposal, current D/Q state and TimingProfile; it returns a StateUpdateProposal plus setup/hold and propagation-delay certificate. All registered state elements sample the same pre-state and commit together, preventing source-code iteration order from changing circuit meaning.

Normative phase schedule

Phase	Meaning
1-3	parse, normalize and resolve content-addressed references
4-6	type, shape and unit validation; dependency closure
7-9	construct candidates; apply declared composition and reduction order
10-12	evaluate exact or bounded results; issue capability and residual certificate

Phase	Meaning
13	support, compatibility and guard classification
14	verified proposal and deterministic conflict resolution
15	atomic commit, lineage, replay and optional projection

Formal VOLT State and Authority Split

q_VOLT = (P, C, N, S, T, M, F, E, L, U) P profile/domains C components and ports N nets/topology S signal and internal state T clocks, time, queue M models/solver profiles F faults/limits E measurements/provenance L waveform/event lineage U residual/uncertainty/status		
Object	Authority role	Boundary
ComponentSpec	Stable ports, parameters, logic/model laws, ratings and provenance.	A symbol or footprint is only a representation.
NetSpec	Stable connectivity relation among compatible ports.	Drawn wire geometry does not define electrical identity alone.
SignalValue	Typed value in a logic or quantitative domain with time and capability.	X, Z, intervals and measured values are not interchangeable.
StateUpdateBatch	Next-state patches sampled from one pre-state at a declared event.	Sequential updates cannot leak source iteration order.
MeasurementRecord	Instrument, calibration, connection, value, uncertainty and provenance.	Simulation output is not a measurement.
TRIPLE AUTHORITY RULE		
A Boolean result may be exact for the formal netlist; a circuit-model result may be bounded for one solver/model; a physical value exists only under the measurement/device evidence profile. These three certificates must not silently promote one another.		

Components, Ports, Nets and Units

```
PortSpec = (id, direction, signal_domain, width, reference_domain,
            drive_or_load_profile, ratings, timing_role, compatibility_tags)
NetSpec = (id, member_ports, resolution_profile, reference_domain, lineage)
```

VOLT typed ports and nets

Visual wires are projections of stable NetSpec records with explicit direction, width and domain.

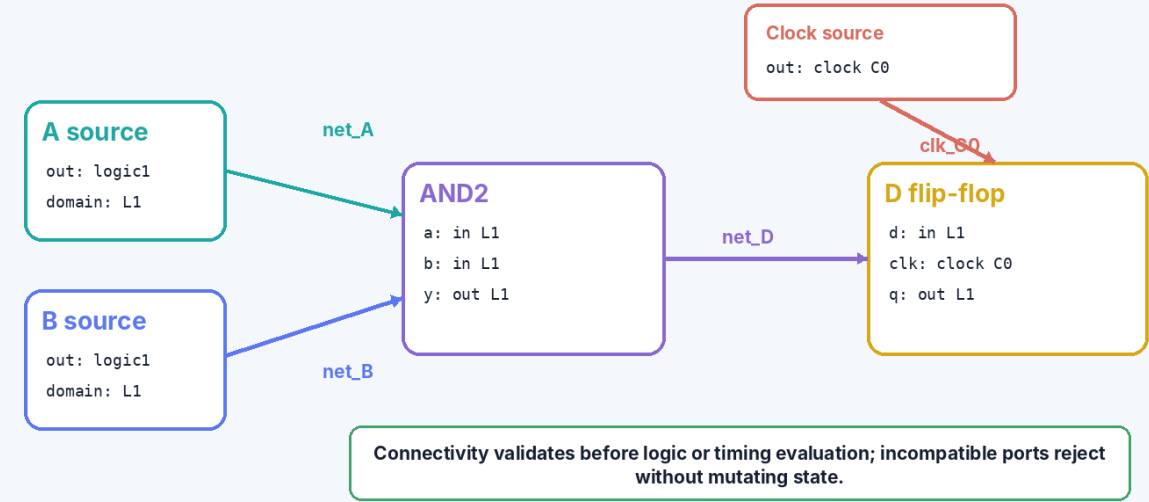


Figure 2. Stable typed ports connect through NetSpec records before logic or timing evaluation.

Check	Example rejection
Direction	two ordinary push-pull outputs connected without a resolution/contention profile
Signal domain	analog voltage port connected directly to a Boolean-only port without an adapter
Width/order	8-bit bus connected to 4-bit port without declared slice/extension and bit order
Reference domain	signal crosses power/reference domain without level/reference adapter
Ratings	profile or authoritative source does not establish compatible limits
Revision	symbol/footprint/port-map version differs from the intended component definition

Units and coercions

Voltage, current, time, charge, resistance, capacitance, inductance, frequency and power retain units and dimensions. Logic thresholds may map quantitative intervals to logic values, but the threshold/profile, hysteresis and uncertainty are explicit operators rather than hidden casts.

Digital Logic and Net Resolution

Value	Meaning in the shipped four-state profile
0	driven or resolved logic low
1	driven or resolved logic high
X	unknown, conflict or uninitialized according to reason metadata
Z	high impedance / no active drive
driver proposals -> enable guards -> stable driver list -> explicit resolution table -> NetValueProposal -> receiving combinational/sequential proposals	
Situation	Required result
one strong 0 driver	resolve 0
one strong 1 driver	resolve 1
all enabled drivers Z	resolve Z or declared pull/default source
conflicting same-strength 0 and 1	resolve X plus contention fault
unknown enable or unknown driver	profile-bound X propagation or conservative rejection
unregistered multiple-driver case	reject netlist/profile
CYCLE BOUNDARY	
Combinational feedback is not silently evaluated in arbitrary component order. A profile must reject it, solve a finite fixed point with certificate, or classify bounded oscillation/unknown status.	

Clock Domains and Sequential Event Order

at time t: 1 collect value/edge proposals 2 resolve nets from q_pre 3 classify active edges and timing guards 4 build all next-state patches from q_pre 5 deterministic conflict/fault policy 6 atomic commit q_post 7 schedule bounded propagation events 8 append waveform and fault lineage	
Timing object	Contract
ClockDomain	source, period/phase or event source, edge set, uncertainty and related-domain policy
Setup/HoldProfile	intervals around active edge plus acceptable input values and ambiguity rule
PropagationDelay	min/max or model-derived delay, transition class and scheduling policy
Delta cycle	zero-physical-time ordering slot with stable queue and convergence/budget rule
Metastability status	timing violation/fault classification; not a physically predictive analog model by default
NONBLOCKING STATE RULE	
All state elements triggered by the same accepted edge sample the same resolved pre-state and commit one batch. A register cannot observe another register's newly written value unless the declared circuit includes a later event.	

Analog and Model Adapter Boundary

Capability	Meaning	Promotion requirement
A0 topology only	components, nets, units and model references resolve	no numerical behavior claim
A1 exact linear symbolic	finite linear relation solved exactly where supported	typed exact arithmetic/profile
A2 bounded numerical	named model and solver produce residual/interval/status	error margin and convergence certificate
A3 calibrated model	parameters tied to measurement/calibration evidence	scope, date, device/lot and uncertainty
A4 observed circuit	physical measurements and conditions accepted into evidence ledger	instrument, calibration, authorization and provenance
A5 governed product	external standards, qualified review and safety/test evidence	authoritative sources and release process
model definition + netlist + initial/boundary conditions -> external solver adapter -> residual / convergence / interval / capability certificate -> optional event proposals -> ordinary support / compatibility / guard / commit		
ORACLE RULE		
Use mature circuit simulators, HDL engines, formal tools and model libraries behind versioned adapters when they are safer or more capable. VOLT is the typed authority and evidence envelope, not a claim to reimplement all of them.		

Faults, Measurements and Physical Authorization

Fault/event	Typical evidence	Authority effect
contention	resolved driver set and strength/profile trace	fault lineage; optional stop/unknown propagation
timing violation	edge, data transition, setup/hold profile and uncertainty	reject or commit fault/unknown according to profile
solver failure	residual, iteration budget, conditioning and terminal reason	no model-based state promotion
over-limit proposal	typed value interval versus authoritative limit/source	reject, derate or fault under policy
measurement accepted	instrument/calibration/connection/value/uncertainty/provenance	evidence patch only; model update needs separate rule
hardware actuation	authorization, interlock, limits, device identity and operator context	physical I/O proposal through governed adapter
SAFETY-CRITICAL BOUNDARY		
Mains, high voltage, batteries, power conversion, medical, automotive, aerospace, industrial control and other hazardous applications require authoritative standards, qualified engineers, protective hardware and physical testing. This cartridge does not supply them.		

Waveform lineage

Waveforms are reconstructed from ordered value-change events, time/delta slots and checkpoints. A screenshot is a useful projection but cannot replace the event record, model profile, uncertainty or measurement provenance.

Executable Notation and OperatorCells

Literal / fallback	Typed meaning	Boundary
AND, OR, XOR, NOT	finite logic operators under a named domain/profile	glyph shape does not choose truth table or unknown propagation
→ / edge arrow	directed signal or transition relation	arrow is not physical causation or delay by itself
~ / AC marker	profile-bound alternating or complement operator	must not overload waveform, negation and approximation
ground/reference glyph	reference-domain port or relation	visual ground symbols do not automatically share one net
clock edge glyph	rising/falling edge guard	needs clock-domain and timing profile
R, C, L labels	component/model family identifiers	physical values/model validity come from exact ComponentSpec and sources
UNICODE / GLYPH RULE		
Foundation 5.0 can co-address literal spelling, canonical glyph and typed semantics [S1]. VOLT uses this for inspectable notation, but no mirrored symbol, local opcode or visual resemblance creates an inverse component, shared reference or electrical fact.		

Deterministic Timing Demonstration

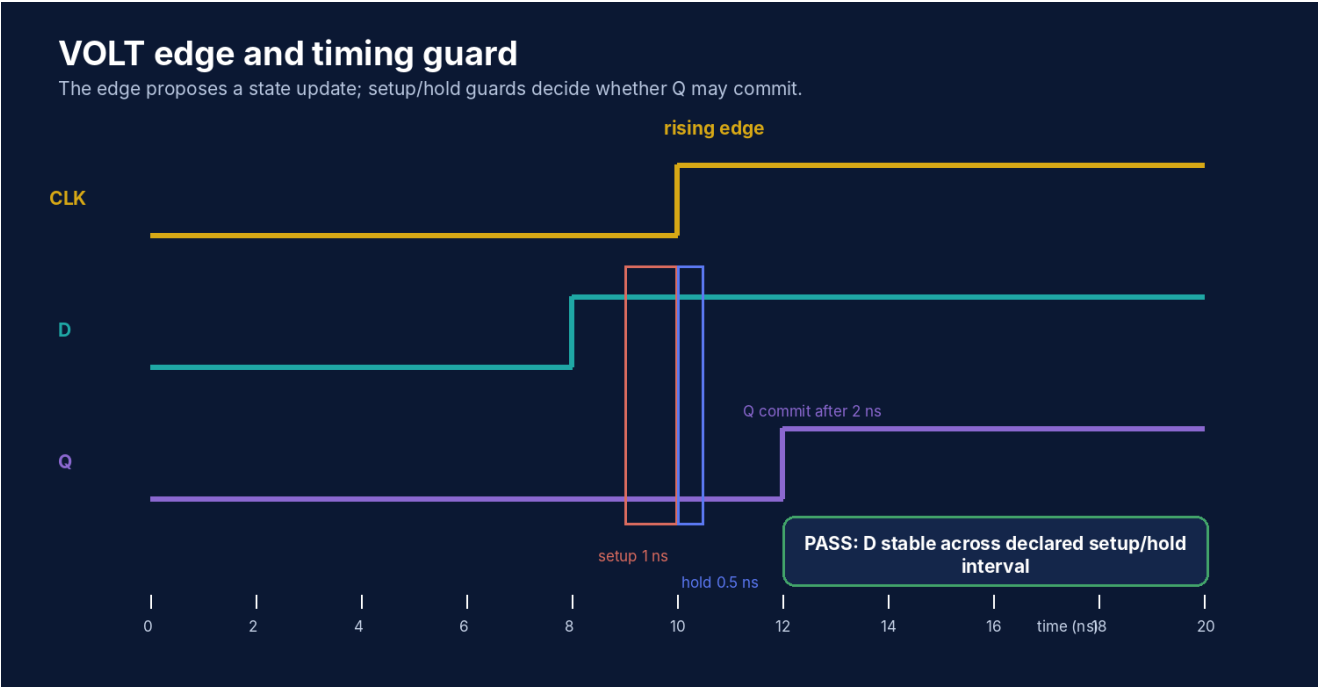


Figure 3. A rising edge becomes a state event only after setup/hold guards pass.

```
profile: setup = 1.0 ns, hold = 0.5 ns, propagation = 2.0 ns
D changes 0 -> 1 at 8.0 ns
CLK rises at 10.0 ns
D remains 1 through 10.5 ns
=> capture proposal PASS; Q commits 0 -> 1 at scheduled 12.0 ns event
```

Fixture	Result	Reason
D stable from 8.0 through 10.5 ns	ACCEPT	setup/hold pass; Q update scheduled
D changes at 9.8 ns	REJECT/FAULT	setup interval violated
D changes at 10.2 ns	REJECT/FAULT	hold interval violated
two strong opposing drivers on D	X + contention	explicit resolution table
clock port connected to logic-domain-incompatible net	REJECT NETLIST	signal-domain mismatch
analog solver returns residual above margin	UNKNOWN / NO PROMOTION	convergence certificate fails

Queries

- `drivers(net_id, time)` → enabled drivers, values, strengths and source lineage
- `resolve(net_id, time)` → value proposal, fault proposal and resolution trace
- `next_events(t0, t1)` → stable ordered edge/delay/fault events
- `timing_check(state_element, edge)` → setup/hold certificate and first failure
- `trace_signal(net_id)` → event waveform with model/measurement capability labels

Packing, Queries and Adapters

Layer	Recommended treatment
Hot digital state	compact logic values, enable flags, queue slots and local IDs with exact codebook
Cold atlas	full component, port, net, model, timing, ratings, sources and hashes
Event stream	time/delta/value/fault changes plus checkpoints; no mandatory dense sample history
Quantities	finite-width code declares unit, range, step, saturation and reconstruction interval
HDL/digital adapter	imports/exports typed netlist and exact finite semantics with unsupported-feature report
SPICE/analog adapter	returns model, solver, residual, convergence and analysis-profile certificate
PCB/bench adapter	downstream design/measurement/actuation with authorization and evidence boundaries
COMPRESSION RULE	
Removing repeated waveform samples can be valid when exact value-change events and time order reconstruct the signal. It is not valid when it discards analog detail, measurement uncertainty, glitches or external novelty required by the selected profile.	

Reference Validation, Boundaries and Kill Criteria

REFERENCE SELF-CHECK RESULT

28/28 deterministic specification self-checks passed. These checks cover catalog uniqueness, required fields, dependency closure, demo arithmetic and declared reason codes. They are package evidence, not independent domain validation.

Checked area	Result
Catalog integrity	48 unique VLT IDs; required fields present.
Connectivity fixture	Port direction/domain/width checks produce deterministic acceptance and reason codes.
Logic resolution	Four-state driver table reproduces 0, 1, Z and X/contention cases.
Timing fixture	Valid capture schedules Q at 12.0 ns; setup/hold violations reject/fault deterministically.
Atomic sequential state	All triggered state elements sample one pre-state in the reference profile.
Model/evidence boundary	High residual and missing calibration fixtures cannot promote physical facts.

Explicit non-claims

- No electrical, EMC, thermal, battery, medical, automotive, industrial or product safety certification is claimed.
- No component values, ratings, layout rules or regulatory requirements are prescribed.
- No analog model or timing number is claimed valid for a physical part without an authoritative source and evidence.
- No HDL, SPICE, formal, synthesis, PCB or lab-tool conformance is claimed.
- No performance/compression advantage is claimed without a named implementation and equal-semantics baseline.

Kill criteria

1. Reject or narrow a use when a symbol, codebook slot or packed record cannot resolve to its exact global definition.
2. Reject when a capability or exactness label is stronger than the model, measurement or external source supports.
3. Reject when units, profiles, coordinate systems, operand order or dependency order are implicit.
4. Reject when a proposal bypasses support, compatibility, guards, deterministic commit or lineage.
5. Reject when an external adapter is required for safety or conformance but its version, calibration or provenance is absent.
6. Reject any performance, compression or safety claim without a named workload, target, error/equivalence contract and evidence.

Complete VOLT 1.0 Mechanism Catalog

The catalog contains 48 cartridge-local mechanisms. These IDs are component-scoped and do not silently extend a global M-number ladder.

ID	Family	Mechanism	Core contract
VLT001	Profile	Circuit profile	Declares digital, mixed-signal or external-analog scope, units, solver and evidence boundary.
VLT002	Identity	Component type	Stable component definition with ports, parameters, laws, models and provenance.
VLT003	Identity	Component instance	Stable instance ID, type/version reference, parameter bindings and lineage.
VLT004	Topology	PortSpec	Direction, signal domain, width, reference domain, ratings and connection policy.
VLT005	Topology	NetSpec	Stable net identity and member ports; wire geometry is downstream representation.
VLT006	Topology	Reference and ground domain	Named reference relation; a ground glyph alone does not create electrical equivalence.
VLT007	Quantity	Voltage quantity	Typed potential difference with unit, reference, interval and uncertainty.
VLT008	Quantity	Current quantity	Typed flow quantity with orientation, unit, interval and uncertainty.
VLT009	Signal	Logic domain	Finite value set, thresholds, unknown/high-impedance semantics and coercions.
VLT010	Signal	Drive strength	Versioned ordering or resolution class for multiple digital drivers.
VLT011	Signal	Impedance/admittance profile	Exact, bounded or oracle-derived relation with frequency/time context.
VLT012	Power	Power-domain profile	Supply, return, limits, sequencing and off-state semantics.
VLT013	Relation	Connect relation	Binds compatible ports to a stable net after width/domain/rating checks.
VLT014	Relation	Driver relation	Declares which outputs may propose net values and under what enable guard.
VLT015	Relation	Load relation	Declares observing ports, input model and fanout/loading contract.
VLT016	Compatibility	Direction compatibility	Rejects output-output, input-input or passive cases unless profile explicitly permits them.
VLT017	Compatibility	Width and bus compatibility	Exact width, slice, concat, extension and endian/order rules are explicit.
VLT018	Compatibility	Electrical-rating compatibility	Voltage/current/power or category checks remain profile- and source-bound.
VLT019	Digital	Combinational operator	Pure typed mapping from current input values to output proposal.
VLT020	Digital	Truth-table or Boolean expression	Finite logic definition with precedence and complete/partial-row policy.
VLT021	Digital	Four-state logic	0, 1, X and Z remain distinct values under a declared logic profile.
VLT022	Digital	Tri-state resolution	Multiple enabled drivers resolve through an explicit table or reject as contention.
VLT023	Digital	Pull/default source	Weak or default drive is a declared component, not hidden simulator behavior.
VLT024	Digital	Combinational cycle profile	Reject, fixed-point solve or bounded-oscillation classification is explicit.
VLT025	Timing	Clock domain	Named clock source, period/phase, edge set, uncertainty and relation to other domains.

ID	Family	Mechanism	Core contract
VLT026	Timing	Edge guard	Classifies rising, falling or level event after time/value prerequisites.
VLT027	Timing	Setup interval	Input must remain acceptable during a declared interval before the active edge.
VLT028	Timing	Hold interval	Input must remain acceptable during a declared interval after the active edge.
VLT029	Timing	Propagation-delay interval	Output update is scheduled inside a declared bounded delay profile.
VLT030	Sequential	State-element transition	Samples inputs and commits next state atomically under edge and timing guards.
VLT031	Analog	KCL constraint	Signed current balance at a node under an explicit model and numerical capability.
VLT032	Analog	KVL constraint	Signed voltage balance around a declared loop under an explicit model.
VLT033	Analog	Linear element model	Resistor/capacitor/inductor or matrix relation with units and time/frequency profile.
VLT034	Analog	Nonlinear model adapter	External device-model oracle with version, parameters, domain and derivative/status contract.
VLT035	Analog	Solver convergence certificate	Residual, iteration/budget, conditioning and terminal status are explicit.
VLT036	Analog	Analysis profile	Operating point, transient, AC or custom bounded analysis with initial/boundary conditions.
VLT037	Authority	Drive proposal	Component output proposes a value, enable state, time and capability to a net.
VLT038	Authority	Net-resolution proposal	Combines current drivers without committing component state.
VLT039	Authority	Atomic state-update batch	Sequential elements update from one sampled pre-state, preventing order leakage.
VLT040	Fault	Fault proposal	Contention, unknown, timing violation, over-limit or solver-failure event with reason code.
VLT041	Evidence	Measurement record	Instrument, calibration, connection, value, uncertainty, timestamp and provenance.
VLT042	Evidence	Calibration/model lineage	Binds simulation and measurements to exact models, sources and versions.
VLT043	Order	Event-queue order	Stable time, delta-cycle, priority and component-ID ordering for deterministic traces.
VLT044	Persistence	Waveform and state lineage	Stores value changes and faults as events rather than ambiguous screenshots.
VLT045	Packing	Compact netlist codebook	Local IDs encode only after exact global definitions and migration records resolve.
VLT046	Adapter	Simulation adapter	External digital/analog simulator returns scoped results and certificates.
VLT047	Adapter	Bench/hardware adapter	Physical I/O remains authorization-, calibration- and safety-gated.
VLT048	Governance	Circuit conformance report	Checks netlist closure, types, timing fixtures, model scope, traces, evidence and non-claims.

Claims Ledger

Claim	Disposition	Reason
Boolean evaluation can be exact.	ADMIT / profile	Exact for a finite declared logic domain and complete operator definitions.
A logic diagram proves a physical circuit works.	REJECT	Physical behavior requires models, parts, power, layout, measurement and evidence.
A wire drawing is a net identity.	REJECT	Stable NetSpec identity and members are authoritative; drawing geometry is downstream.
Multiple drivers resolve automatically.	REJECT	Resolution or contention requires an explicit driver/strength/profile table.
Setup/hold checks prove hardware timing.	BOUNDED	They prove only the declared timing contract; physical values require authoritative sources and measurement.
Four-state X is a probability.	REJECT	X denotes unknown/conflict under a logic profile, not a numeric probability.
Ground symbols are universally equivalent.	REJECT	Reference domains and connection relations must be explicit.
Analog convergence proves the model is true.	REJECT	Convergence means a solver met its numerical contract for the chosen model.
A seed reconstructs measured waveforms.	REJECT	External novelty and measurement evidence must be stored.
A compact netlist is lossless.	ADMIT / contract	Only when decode preserves exact topology, definitions, order and required error metadata.
VOLT supplies electrical safety certification.	REJECT	No code, standard, qualified design review or lab test is imported.
VOLT replaces SPICE or HDL tools.	REJECT	Mature simulators and synthesis/formal tools should remain external adapters for their scope.

High-Impact Application Profiles

Profile	What the cartridge enables
Circuit-building game and learning lab	Typed ports physically refuse incompatible connections and explain the first failed rule.
Deterministic digital simulator	Exact finite logic, explicit X/Z and event-order replay without hidden scheduler behavior.
Proof-carrying interface checker	Validate revisions, buses, clock domains, level/reference adapters and timing contracts before downstream tooling.
Mixed-tool digital thread	Connect HDL, formal, SPICE, PCB and metrology outputs through scoped adapters and certificates.
Bench evidence ledger	Record calibration, measurements, faults and model-vs-observation residuals offline and replayably.

Final Formal Definition

FORMAL DEFINITION

UGTOMS VOLT 1.0 is a component-scoped circuit and logic cartridge in which component identity, typed ports and nets, units, finite logic values, driver resolution, clock/timing guards, sequential state, model profiles, faults, measurements and waveform lineage are literal substrate data; digital, model and physical certificates remain distinct; and every state or hardware effect passes through the retained proposal, verification, deterministic commit and evidence chain.

Implementation sequence

7. Freeze schemas for ComponentSpec, PortSpec, NetSpec, TimingProfile, MeasurementRecord and CircuitCertificate.
8. Implement the finite four-state logic, resolution table, event queue and atomic state-batch scalar oracle.
9. Ship the AND-to-DFF timing fixture plus contention and setup/hold rejection cases.
10. Add read-only HDL and SPICE adapters with explicit unsupported-feature and residual reports.
11. Require qualified electrical sources, review and physical tests before any hardware or safety promotion.

RELEASE CONCLUSION

This 1.0 document freezes a domain-facing contract. It is deliberately useful before a production engine exists: implementations can now disagree visibly through typed profiles, reason codes and certificates instead of hiding differences behind the same symbol.